

CDC X Yhills OPEN PROJECTS 2025-2026

Data Science Project

Arnav Agarwal 23118009

Overview

This project aims to predict property prices by leveraging both tabular data and features extracted from satellite images. The workflow integrates traditional machine learning models with deep learning techniques to enhance prediction accuracy.

Initially, various regression models including Linear, Ridge, Lasso, Decision Tree, Random Forest, and Gradient Boosting are trained and evaluated using only the tabular property data to establish baseline performance. To enhance predictive power, satellite images corresponding to each property are processed to extract features through both classical image processing techniques and deep learning methods, specifically convolutional neural networks (CNNs).

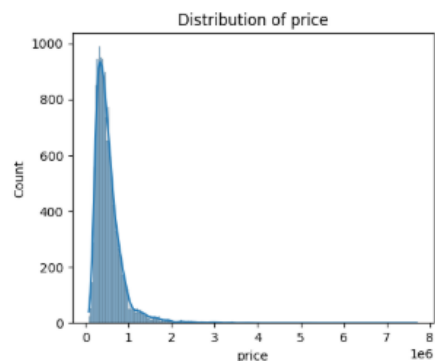
These image-derived features are then merged with the original tabular dataset. Further, feature engineering is performed by applying Principal Component Analysis (PCA) to reduce the dimensionality of the CNN features, and by incorporating engineered metrics such as green area ratio and concrete area ratio. Finally, Gradient Boosting models are retrained on this enriched dataset, allowing for a comprehensive assessment of the performance improvements gained by combining tabular and image-based features.

Exploratory Data Analysis

The exploratory data analysis (EDA) phase of the project involved a comprehensive examination of the property dataset to uncover underlying patterns, trends, and relationships among variables.

1. Price Distribution

The price distribution analysis reveals how property prices are spread across the dataset. By visualizing the distribution using histograms, we can identify the central tendency, detect skewness, and spot potential outliers. This helps in understanding the overall market range and informs further modelling decisions.



2. Satellite Images

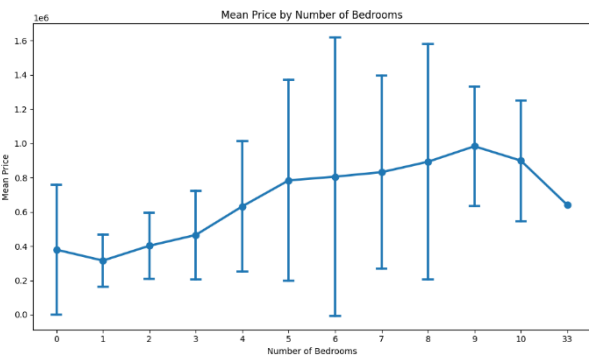


The satellite images used in this project are 512x512 pixels in size, each centered on the property's location. This resolution covers an area of approximately 220 by 220 meters, offering extensive coverage of the property's surroundings. Such a wide field of view provides rich contextual information about the neighborhood and environment, enabling the neural network to make more informed and accurate inferences based on both the property and its immediate vicinity.

Financial Insight

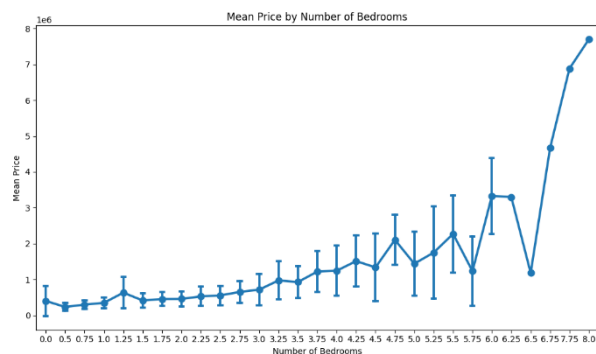
1. Impact of No. of Bedrooms in Price

As the number of bedrooms increases, the average price of houses also tends to rise. This trend indicates that properties with more bedrooms generally command higher prices, reflecting the added space and utility they offer to potential buyers.



2. Impact of No. of Bathrooms in Price

A similar trend is observed with the number of bathrooms: as the number of bathrooms in a house increases, the average price of the property also rises.



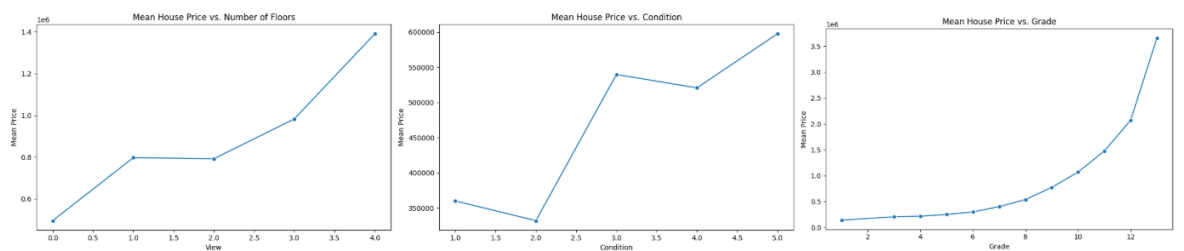
3. Impact of Living Area to Price

The scatter plot clearly demonstrates a positive relationship between the two variables. As the living area of a property increases, the price generally rises as well. This trend suggests that larger homes, offering more space and utility, are valued higher in the market.



4. Impact of other columns on Price

A similar pattern is observed in many other features, such as condition and grade, where higher values are directly correlated with increased house prices.



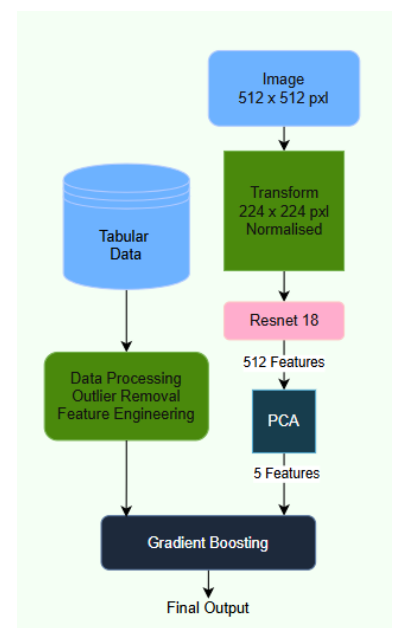
Model Architecture

Image Features: Obtained from pretrained Resnet18

Image Extracted Features: Green Area ratio and Concrete Area ratio obtained from the images using OpenCV

Multimodal Model Architecture:

The model uses a dual-input architecture combining image and tabular data. Images are resized to 224x224, normalized, and passed through a ResNet-18 backbone to extract 512 deep features, which are further reduced to 5 dimensions using PCA. In parallel, tabular data undergoes preprocessing, including outlier removal and feature engineering. The processed features from both branches are fused and fed into a Gradient Boosting model to produce the final prediction.



Results

1. Tabular Data

Selection of Regression Model

Model	R2 Score	RMSE	MAE	MAPE (%)
Linear Regression	0.71	191571	125410.4	25.84
Ridge Regression	0.71	191613.7	125386.3	25.83
Lasso Regression	0.71	191571.8	125409.3	25.84
Decision Tree	0.73	182398.6	102293.9	18.68
Random Forest Tree	0.86	131819.1	72196.87	13.19
Gradient Boosting	0.86	130178.8	78291.58	14.73

Gradient Boosting was chosen over Random Forest tree as gradient boosting focuses on reducing bias and can capture complex patterns more effectively, especially in structured/tabular data. Random Forest is more prone to overfitting as compared to gradient boosting.

Gradient Boosting Metrics after Hyperparameter tuning

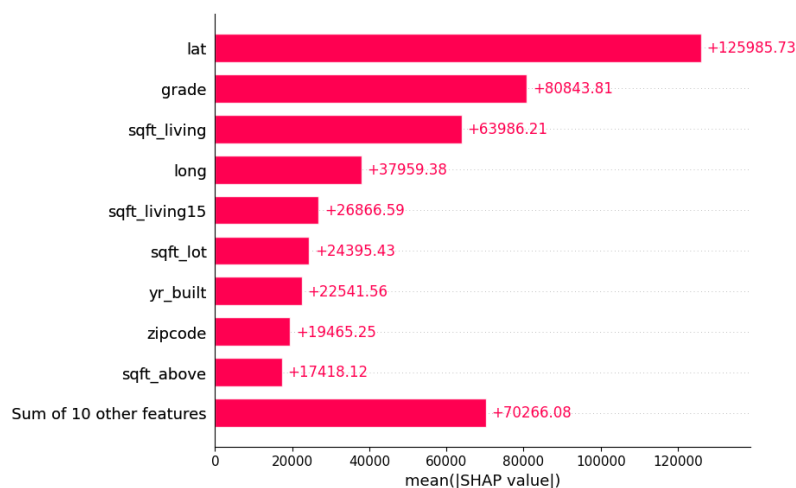
R2=0.8900, RMSE=118641.97, MAE=69378.72, MAPE=12.98%

2. Tabular Data + Image Features + Image Extracted Features

Model	R2 Score	RMSE	MAE	MAPE (%)
CNN Only	0.4713	257564.69	190161.02	40.91
Gradient Boosting Only	0.8900	118641.97	69378.72	12.98
Gradient Boosting + CNN Features	0.9000	114174.37	68400.27	12.91

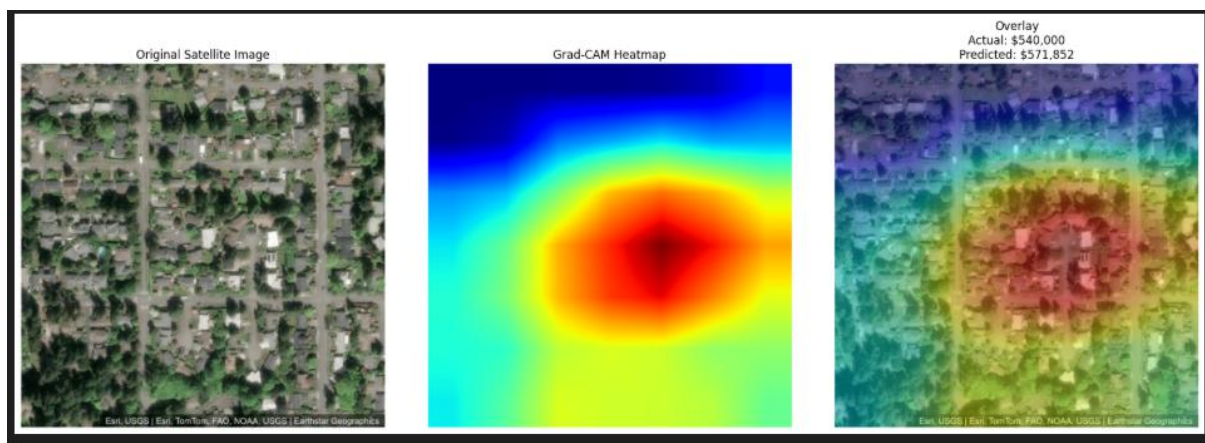
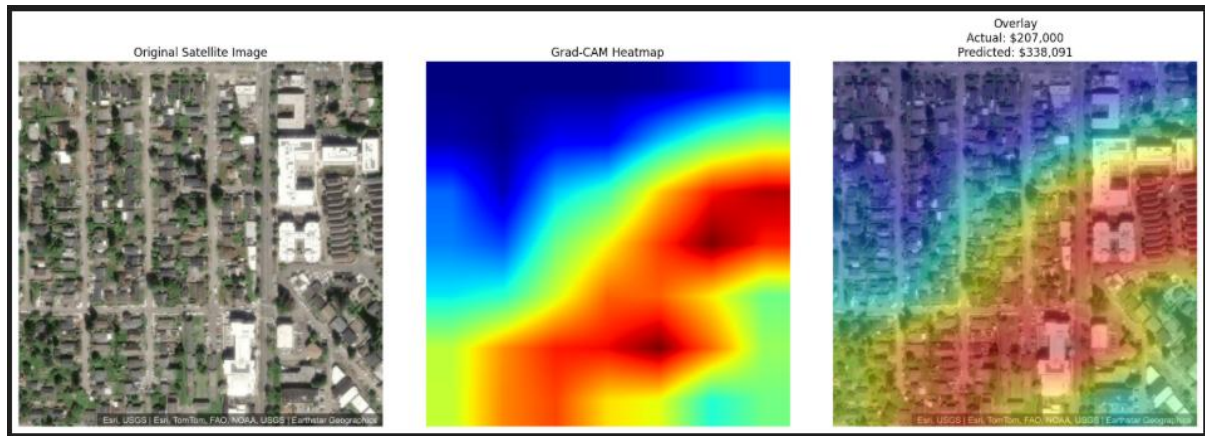
Model Explainability

1. SHAP (For Gradient Boosting)



These are the features which contributed the most when predicting using Gradient Boosting.

2. GradCAM (For CNN)



In the provided Grad-CAM visualizations, it is evident that the regions surrounding the property, particularly the center of the satellite image where the property is typically located, exhibit more intense and dense coloration. This denser color in the heatmap indicates that these areas have a greater influence on the model's price prediction for that specific property.

In the context of property price prediction, the immediate surroundings such as the presence of green spaces, roads, water bodies, or dense urban infrastructure can significantly affect the perceived value of a property.